



MECH MAINT- TURBINE Department
SIMHADRI

Ref:SMTTP/MECH MAINT-/2022-23/MM/TMD/110344

Date:

18/05/2022(11:22:11)

Subject:Revision of LMI Safety precautions to be taken when carrying out inspection and maintenance of turbine componentsreg

The LMI: “**Safety Precautions to be taken when Carrying out Insp. And Maint. On Turbine Gen. Component which are not Completely Self Supporting**” LMI NO. LMI/OGN/OPS/MECH/023” is revised by CC-OS.

Following changes were done in LMI as per revised OGN:

- **Point no-7a:** OEM fixtures usage clause added
- **Point no-7g:** OEM supports usage clause added

Put up for Approval of Competent authority.

Encl:

- Earlier approved LMI.
- New LMI
- Revised OGN

Submitted by:

रामबाबू वानापल्ली - उप महाप्रबंधक

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On:18/05/2022(11:22:11) **Ref:**SMTTP/MECH MAINT-/2022-23/MM/TMD/110344

Forwarded for approval please.

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

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On:18/05/2022(17:40:15) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

forwarded for further approval

Submitted by:

आई एल रेड्डी - अपर महाप्रबंधक

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Submitted for approval of competent authority

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On:18/05/2022(17:48:45) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

The LMI attached in the Annexures is '**LMI/OGN/OPS/MECH/021**' i.e. for 'GUIDANCE FOR THE REMOVAL OF SIEZED STUDS BY THE DRILLING/BORE WELDING METHOD' whereas the LMI under approval is '**LMI/OGN/OPS/MECH/023**' for 'SAFETY PRECAUTIONS TO BE TAKEN WHEN CARRYING OUT INSPECTION AND MAINTENANCE ON TURBINE GENERATOR COMPONENTS WHICH ARE NOT COMPLETELY SELF SUPPORTING.'

Please attach the relevant LMI copy(old - unrevised) that is to be revised, for reference.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:19/05/2022(15:58:11) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

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On:19/05/2022(18:09:39) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/110344

Submitted by:

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On:19/05/2022(18:19:46) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/110344

Reviewed.

Submitted by:

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On:20/05/2022(09:17:09) **Ref:**SMTPP/MECH MAINT-/2022-23/MM/TMD/110344

Needful done.

May pl be approved

Submitted by:

जीजीके मूर्ति - अपर महाप्रबंधक

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On:20/05/2022(09:33:03) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

Kindly review Cl. no. 9.0 in Revised LMI, as discussed.

Submitted by:

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On:21/05/2022(10:16:36) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

Please review as discussed.

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

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On:25/05/2022(10:01:00) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

Corrections done.

Submitted by:

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On:25/05/2022(10:45:50) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

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On:31/05/2022(19:25:54) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

Revised LMI may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:08/06/2022(18:40:42) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

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On:09/06/2022(20:07:42) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

Approved & Submitted by:

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On:10/06/2022(09:14:17) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/110344

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/023/

ISSUE NO.1; REVISION NO. 04; DATE: MAY'2022

**SAFETY PRECAUTIONS TO BE
TAKEN WHEN CARRYING OUT
INSPECTION AND MAINTENANCE
ON TURBINE GENERATOR
COMPONENTS WHICH ARE NOT
COMPLETELY SELF SUPPORTING**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	SAFETY PRECAUTIONS TO BE TAKEN WHEN CARRYING OUT INSPECTION & MAINTENANCE ON TURBINE GENERATOR COMPONENTS WHICH ARE NOT COMPLETELY SELF SUPPORTING
<u>2.a) DOCUMENT CODE</u> b) REVISION/ISSUE NO.&DATE	<u>LMI/OGN/OPS/MECH/023/</u> <u>ISSUE-1; REV.-04; MAY'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/023</u>
<u>4.PRERPARRED BY</u>	DGM (TMD)
<u>5. CHECKED BY</u>	AGM (TMD)
<u>6.RECHECKED BY</u>	GM (MAINT)
<u>7. REVIEWED BY</u>	GM (O & M)
<u>8. APPROVED BY</u>	HOP
<u>9.INDEX LIST</u>	ENCLOSED.
<u>10.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: SAFETY PRECAUTIONS TO BE TAKEN WHEN CARRYING OUT INSPECTION AND MAINTENANCE ON TURBINE GENERATOR COMPONENTS WHICH ARE NOT COMPLETELY SELF SUPPORTING		
REV:04	DATE: MAY'2022	
REF NO. LMI/OGN/OPS/MECH/023/		PAGE : 1 OF 7
<u>INDEX</u>		
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2.0	INTRODUCTION	2
3.0	SCOPE	2
4.0	REFERENCE DOCUMENT	2
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6.0	SPECIFIC INSTRUCTIONS	3
7.0	METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS	3
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NTPC	LMI	SIMHADRI
TITLE: SAFETY PRECAUTIONS TO BE TAKEN WHEN CARRYING OUT INSPECTION AND MAINTENANCE ON TURBINE GENERATOR COMPONENTS WHICH ARE NOT COMPLETELY SELF SUPPORTING REV:04 DATE: MAY'2022		
REF NO. LMI/OGN/OPS/MECH/023/		PAGE : 2 OF 7
<u>2.0 INTRODUCTION</u> At Simhadri the steam turbine is a tandem compound machine with separate HP,IP and LP sections. The HP turbine is a single flow machine with its outer casing being of barrel type without axial or radial flange. The inner casing of HP turbine which is axially split is arranged in the barrel type outer casing. The IP turbine is of double shell construction with horizontally split casing. A double flow inner casing is supported in outer casing. The double flow LP turbine is of three shell construction which are horizontally split. The inner most shell which carries the front guide blade rows is suspended in the intermediate shell to allow for thermal expansion. The intermediate shell carries the rear guide blade rows and is supported in the outer casing in a manner permitting free radial expansion, concentric with the shaft and axial expansion from a fixed point. The outer casing consists of front and rear walls and is supported by ends of longitudinal beams on the base plates of foundation. There have been reported incidents of fatal accidents due to diaphragms from LP turbine top half cylinders in set down condition falling onto personnel working below. The objective of this LMI is to regiment a framework of guidelines for preventing unsafe conditions during inspection and maintenance of turbine generator. <u>SCOPE</u> The scope of this LMI is to specify the mandatory norms to be adhered to when carrying out inspection and maintenance on turbine generator components which are not completely self supporting. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note OD/OGN/MECH/023 circulated by operation services- NTPC corporate center, Rev: 01 <u>5.0 SUPERSEDED DOCUMENT</u> LMI/OGN/OPS/MECH/023 REV 03, DATE: FEB'2022		

NTPC	LMI	SIMHADRI
TITLE: SAFETY PRECAUTIONS TO BE TAKEN WHEN CARRYING OUT INSPECTION AND MAINTENANCE ON TURBINE GENERATOR COMPONENTS WHICH ARE NOT COMPLETELY SELF-SUPPORTING. REV:04 DATE: MAY'2022		
REF NO. LMI/OGN/OPS/MECH/023/		PAGE : 3 OF 7
<u>6.0 SPECIFIC INSTRUCTIONS</u> The objective of this LMI is sought to be achieved by a) adherence to safety precautions while carrying out inspection and maintenance on TG components. <u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u> a) When top half cylinders are being lifted, complete with gland carriers, ensure designed locking arrangement is in position. Specially designed fixtures provided by OEM shall be used wherever required. b) All personnel shall be kept away during transportation of suspended components by like casing half cylinders and turbine/generator rotors by overhead crane. c) Inspection and maintenance work shall not be carried out beneath a suspended component. No person shall be allowed to pass under a suspended component. d) if there is a good reason for carrying inspection/ cleaning work beneath a turbine cylinder in its assembled condition, suitable , adequate and purpose designed steel supports/ timber baulks, for the cylinder and back up supports to guard against the failure of holding devices of sub-components within the cylinder shall be provided. e) Load testing of EOT cranes, slings used for components should be ensured before the above operation. f) If major work is required within the top half cylinder, such as removal / replacement of gland carriers, the half cylinder complete with assembled components shall be inverted and set down on suitable supports before work commences.		

NTPC	LMI	SIMHADRI						
TITLE: SAFETY PRECAUTIONS TO BE TAKEN WHEN CARRYING OUT INSPECTION AND MAINTENANCE ON TURBINE GENERATOR COMPONENTS WHICH ARE NOT COMPLETELY SELF SUPPORTING								
REV:04 DATE: MAY'2022								
REF NO. LMI/OGN/OPS/MECH/023/		PAGE : 4 OF 7						
g) Purpose designed support stands or those supplied by OEM must be used while placing turbine components after removal from position h) Any difficulty in adhering to above precautions shall be referred to OEM . RESPONSIBILITY- HOD (TMD) for all above points <u>8. EXCEPTION REPORTING</u> Any exceptions to the instructions specified in the LMI and incidence of failure of holding device of any sub-components of turbine cylinders while carrying out inspection and maintenance reported as per the following format to HOD (MM) Responsibility- HOD (TMD) <table border="1"> <tr> <th>Description of exception</th> <th>Unit No.</th> <th>Remarks</th> </tr> <tr> <td> </td> <td> </td> <td> </td> </tr> </table> <u>9.0 RESPONSIBILITIES</u> The responsibility of carrying out different tasks as detailed in the LMI are with HOD (TMD). <u>10. RECORD KEEPING</u> The records to be maintained under this LMI include a) Compliance record of the activities detailed under this LMI b) Exception reporting The above records are to be maintained by MM – Turbine Maintenance <u>11.REVIEW</u> This document shall be reviewed by HOP, SIMHADRI on two-yearly basis.			Description of exception	Unit No.	Remarks			
Description of exception	Unit No.	Remarks						

[illegible]

[illegible]

NTPC	COMPLIANCE RECORD	SIMHADRI
TITLE: SAFETY PRECAUTIONS TO BE TAKEN WHEN CARYING OUT INSPECTION AND MAINTENANCE ON TURBINE GENERATOR COMPONENTS WHICH ARE NOT COMPLETELY SELF SUPPORTING REV:04 DATE: MAY'2022		
REF NO. LMI/OGN/OPS/MECH/023/		PAGE : 7 OF 7
UNIT NO/Date	ACTIVITY	COMPLIANCE